

8. (3.75 points)

Given the following information:

- Two valuation classes, home and auto, are in runoff.
- Outstanding claims liabilities: \$45 million for home and \$5 million for auto.
- Internal systemic risk was analyzed using the following balanced scorecard assessment:

Risk Component	Home Weight	Auto Weight
Specification Error	5	2
Parameter Selection Error	3	5
Data Error	2	3

Potential Risk Indicators (Worst = 1, Best = 5)	Home Score	Auto Score
a. Best predictors are stable over time or change due to process changes	5	2
b. Knowledge of past processes affecting predictors	2	2
c. Number of independent models used	3	5
d. Extent, timeliness, consistency and reliability of information from business	1	3
e. Best predictors have been identified, whether or not they are used	4	1
f. Range of results produced by models	3	5

Score from Balanced Scorecard Assessment	Home Coefficient of Variation	Auto Coefficient of Variation
1.0 to 2.0	8.5%	6.5%
2.0 to 3.0	7.0%	5.0%
3.0 to 4.0	6.0%	4.0%
4.0 to 5.0	5.5%	3.5%

- Internal systemic risk has a correlation coefficient of 0.40 between valuation classes.
- Total independent risk coefficient of variation is 4.0%
- Total external systemic risk coefficient of variation is 8.0%
- The underlying distribution of insurance liabilities is Normal.
- The 80th percentile of the Standard Normal distribution is 0.842

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(8 continued)

a. (0.5 point)

Describe a scenario that could justify a score of 1 for the following potential risk indicator:

Extent, timeliness, consistency and reliability of information from business

b. (0.75 point)

Assign each potential risk indicator to the appropriate risk component.

c. (0.5 point)

Calculate the home and auto weighted average balanced scorecard assessment scores, using the potential risk indicator assignments from part b. above.

d. (1 point)

Calculate the total internal systemic risk coefficient of variation.

e. (1 point)

Calculate the risk margin in dollars for a required probability of adequacy of 80%.

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QUESTION 8	
TOTAL POINT VALUE: 3.75	LEARNING OBJECTIVE(S): A5, A8
SAMPLE ANSWERS	
Part a: 0.5 point	
<u>Sample 1</u> A score of 1 would indicate worst practice. A possible scenario for this score might include: - highly summarized reports with little/no detail - mislabeled or poorly identified data - errors in calculations - missing data without explanation clearly given and/or - irregular, sporadic, or missing reports (for example, no January report given) <u>Sample 2</u> Very little or infrequent communication between actuarial team and claim department - Information not up to date - poor timeliness and reliability of information <u>Sample 3</u> Data is on an extreme lag, sometimes unavailable, and audits regularly find incorrect and missing data which isn't accounted for	
Part b: 0.75 point	
A = Parameter selection error B = Data error C = Specification error D = Data error E = Parameter selection error F = Specification error	
Part c: 0.5 point	
<u>Sample 1</u> Home Avg BSC = $(5*3 + 2*2 + 3*5 + 1*2 + 4*3 + 3*5)/(3 + 2 + 5 + 2 + 3 + 5) = 3.15$ Auto Avg BSC = $(2*5 + 2*3 + 5*2 + 3*3 + 1*5 + 5*2)/(5 + 3 + 2 + 3 + 5 + 2) = 2.5$ <u>Sample 2</u> Home: Specification = $(3+3)/2 = 3$ Parameter Selection = $(5+4)/2 = 4.5$ Data = $(2+1)/2 = 1.5$ Overall = $[5(3) + 3(4.5) + 2(1.5)] / (5+3+2) = 3.15$ Auto: Specification = $(5+5)/2 = 5$ Parameter Selection = $(2+1)/2 = 1.5$ Data = $(2+3)/2 = 2.5$	

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$$\text{Overall} = [2(5) + 5(1.5) + 3(2.5)] / (2+5+3) = 2.5$$

Part d: 1 point

Sample 1

Home CoV = 6%

Auto CoV = 5%

$$w(\text{home}) = 45/50 = .9$$

$$w(\text{auto}) = 5/50 = .1$$

$$\text{Internal CoV} = (.9^2 * .06^2 * .1^2 * .05^2 + .4^2 * .9 * .1 * .06 * .05)^{0.5} = 0.56$$

Sample 2

$$\phi(\text{Home}) = 6.0\%$$

$$\phi(\text{Auto}) = 5.0\%$$

$$w(\text{Home}) = 45/50 = .9$$

$$w(\text{Auto}) = 5/50 = .1$$

$$\Sigma = \begin{pmatrix} 1.0 & .40 \\ .40 & 1.0 \end{pmatrix}$$

Using Matrix Multiplication

$$(\phi w) = (.054 \ .055)$$

$$(\phi w) \Sigma = (.056 \ .0266)$$

$$\begin{aligned} \phi^2(\text{Internal}) &= (\phi w) \Sigma (\phi w) \\ &= .054 * .056 + .005 * .0266 = .003157 \end{aligned}$$

$$\phi(\text{Internal}) = .0562$$

Part e: 1 point

Sample 1

$$\text{Risk Margin} = \phi z \mu = (.056^2 + .04^2 + .08^2)^{0.5} * .842 * (50M) = \$4.44$$

EXAMINER'S REPORT

Candidates were expected to be able to demonstrate their knowledge of various sources of risk and uncertainty, and how scores are determined in a Balanced Scorecard Assessment. Candidates were also expected to calculate risk margins that consider these sources of risk and uncertainty.

Part a

Candidates were expected to identify a scenario that has a higher risk potential and is related to the Data Error risk component.

Common errors include:

- Identifying a risk that was not a Data Error risk, such as risk of adverse development or cat risk.
- Identifying a scenario that has a lower risk potential, such as data that was provided in an accurate or timely manner.
- Defining Data Error risk without identifying a scenario that would result in a higher risk

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potential.
Part b
Candidates were expected to classify potential risk indicators into three possible risk components. Common errors include: • Categorizing item b (Knowledge of past processes affecting predictors) as a Parameter Selection Error. • Categorizing item b (Knowledge of past processes affecting predictors) as a Specification Error.
Part c
Candidates were expected to calculate balanced scorecard assessment scores, based on their selections in part b., the home and auto weights for each of the 3 risk components, and the home and auto scores for each of the 6 potential risk indicators. Common errors include: • Using separate CoV lookups for each of the 3 risk components and calculating their weighted average. • Using a weight of (5+3+2) in the denominator after having selected more than 2 of any risk component in part b. or using the same weights for both auto and home.
Part d
Candidates were expected to look up the home and auto CoVs based on the balanced scorecard assessment scores from part c., calculate the home and auto weights based on outstanding claims liabilities, and compute the internal systemic risk CoV. Common errors include: • Not using the balanced scorecard assessment scores from part c. to look up the correct CoV from the table. • Using 50/50 weighting between home and auto instead of 90/10. • Not squaring the $w(i) \cdot \text{Cov}(i)$ terms in the internal systemic risk calculation. • Omitting the $2 \cdot \rho \cdot w(\text{home}) \cdot w(\text{auto}) \cdot \text{Cov}(\text{home}) \cdot \text{Cov}(\text{auto})$ in the internal systemic risk calculation. • Not multiplying the $\square \cdot w(\text{home}) \cdot w(\text{auto}) \cdot \text{Cov}(\text{home}) \cdot \text{Cov}(\text{auto})$ by 2 in the internal systemic risk calculation. • Not using the square root in the internal systemic risk calculation.
Part e
Candidates were expected to calculate the total insurance liabilities CoV, based on the total independent risk CoV, the external systemic risk CoV, and the internal systemic risk calculation calculated in part d. Candidates were also expected to calculate the risk margin based on the total insurance liabilities CoV, the 80th percentile of the Standard Normal distribution, and the dollar amount of the outstanding claims liabilities. Common errors include: • Neglecting to calculate the total insurance liabilities CoV, and instead use the total internal systemic risk coefficient for the calculation of the risk margin. • Not squaring the terms in the calculation of the total insurance liabilities CoV.

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- Not using the square root in the calculation of the total insurance liabilities CoV.
- Not converting the risk margin to dollars.
- Adding \$50 million to the risk margin.